The Logit Normal Distribution

Wyatt Clegg

December 8, 2016

1 Introduction

The Logit-Normal Distribution is the occult cousin of the normal distribution. It appears in compositional data analysis, or as a substitute when we don't want to use a Dirichlet distribution.

1.1 History

From an examination of the literature, it appears that Dr. John Aitchinson and Dr. S. M. Shen of the University of Hong Kong developed the distribution in 1980.[1] Initially, they proposed using it for compositional and proportional data. It was later found that the Logit-Normal is a good substitue for the Dirichlet distribution.[3]

Compositional data was the main emphasis of Aitchinson's original paper. Therefore, I selected milk fat composition data from a data set used previously to evaluate the bootstrap method as my test data[2].

Potentially, an increased understanding of the distribution of milk fat composition could help us drain cream off milk more efficiently. We could then produce milk to fit sales specifications for fat composition more efficiently and reliably.

2 Distribution description

The original pdf, as defined by Aitchinson[1], is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}x(1-x)} \exp\left(-\frac{1}{2}\left[\frac{\frac{x}{1-x}-\mu}{\sigma}\right]^2\right) \text{ for } x \in (0,1)$$

While a closed form CDF exists, this classification is a misnomer. The CDF includes an Error Function, which by definition is not closed form. Therefore, I didn't decide to include it, because for use as a closed form formula it is useless.

3 Parameters

The Logit-Normal distribution has two parameters: the location μ and scale σ^2 parameters. These correlate to the original normal distribution that we transformed our data from, such that for $X \sim LogitN(\mu, \sigma^2)$,

$$\text{logit}(X) \sim N(\mu, \sigma^2).$$

In the normal distribution, μ and σ^2 correspond to the mean and variance, respectively. This relation does not hold up in the Logit-Normal distribution.

The location parameter tells us roughly where the mass is centered about. If μ is close to 0, the distribution centers around $\frac{1}{2}$. If μ is a large positive number, the mass of the distribution gravitates close to 1. Similarly, the mass of the distribution moves towards 0 when μ is a large negative number.

The scale parameter is even less interpretable. It roughly tells us how far the mass is pushed away from the center of our distribution. Neither μ nor σ^2 translate into anything usable.

3.1 Derivation of the Maximum Likelihood Estimators

The original pdf, as defined earlier, is

$$f(x) = \tfrac{1}{\sigma\sqrt{2\pi}x(1-x)} \exp\left(-\tfrac{1}{2}\left[\tfrac{\text{logit}(x)-\mu}{\sigma}\right]^2\right) \text{ for } x \in (0,1)$$

The likelihood function is then

$$L(X) = \prod_{i=1}^{n} f(x_i) = \left(\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\mu^2}{2\sigma^2}\right)\right)^n \prod_{i=1}^{n} \frac{1}{x_i(1-x_i)} \exp\left[-\frac{\text{logit}^2(x)}{2\sigma^2} + \frac{\mu\,\text{logit}(x)}{\sigma^2}\right]$$

Further, we define the log-likelihood as

$$l(X) = -n\log(\sigma) - \frac{n}{2}\log 2\pi - \frac{n\mu^2}{2\sigma^2} - \sum_{i=1}^{n}\log(x_i(1-x_i)) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\text{logit}^2(x_i) + \frac{1}{\sigma}\sum_{i=1}^{n}\mu(\text{logit}(x_i)).$$

From here we can solve for the Maximum Likelihood Estimators (MLE's) by taking the derivative relative to the parameters, setting it equal to 0, and solving for the parameter.

3.2 MLE μ

First we take the derivative of the likelihood function relative to μ.

$$\frac{d}{d\mu} l(X) = \frac{2n\mu}{2\sigma^2} + \frac{1}{\sigma^2} \sum_{i=1}^{n} \text{logit}(x_i) = 0$$

$$\frac{n\mu}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \text{logit}(x_i)$$

$$n\mu = \sum_{i=1}^{n} \text{logit}(x_i)$$

$$\mu_{mle} = \frac{\sum_{i=1}^{n} \text{logit}\, x_i}{n}$$

3.3 MLE σ^2

Notice that when we declared the log-likelihood originally, we simplified a certain way to make the derivation more accessible. Observe that an alternate way to write the log-likelihood is

$$l(X) = -\frac{n}{2} \log(\sigma^2) - \frac{n}{2} \log(2\pi) - \sum_{i=1}^{n} \log(x_i(1 - x_i)) \frac{1}{2\sigma^2} \sum_{i=1}^{n} (\text{logit}(x_i) - \mu)^2$$

So if we take the derivative with respect to σ^2 we get

$$\frac{d}{d\sigma^2} l(X) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (\text{logit}(x_i) - \mu)^2$$

$$n = \frac{1}{\sigma^2} \sum_{i=1}^{n} (\text{logit}(x_i) - \mu)^2$$

$$\sigma^2_{mle} = \frac{1}{n} \sum_{i=1}^{n} (\text{logit}(x_i) - \mu)^2.$$

These two measures constitute our Maximum Likelihood estimators for our location μ and shape σ^2 parameters, respectively.

4 Other Estimators

I chose to construct two Bayesian estimators as alternatives to the MLE. While this seems to lack imagination, I hope to make my estimators interesting nonetheless.

4.1 Bayes I

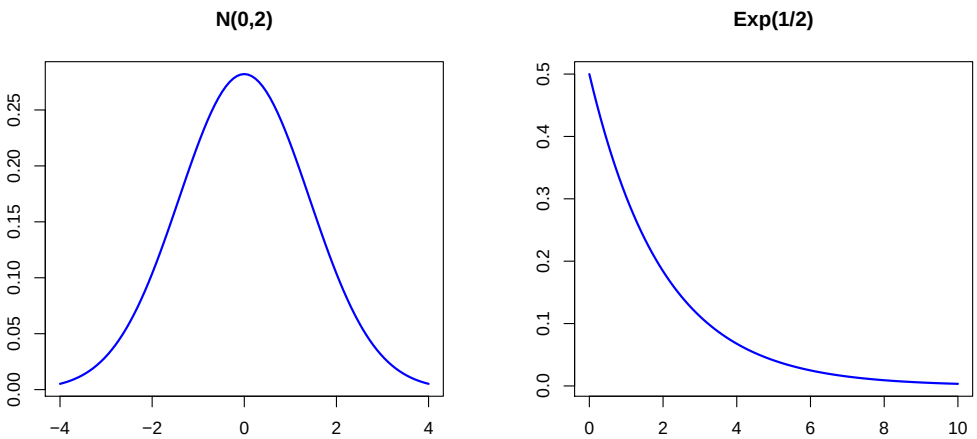
My first Bayesian estimator will be less informed. I want to contrast this and my second, more informed Bayesian estimator. Despite this goal, the prior distribution for μ is identical for each.

This study will show how a far more informative prior for σ^2 will change estimates of μ and impede parameter estimation overall. Also, I will compare use of the Exponential versus the Inverse Gamma distribution as priors on σ^2.

I chose, as a prior distribution that $\mu \sim N(0, 2)$. The relatively large variance is a largely uniformative prior even though it carries some weight with it.

For σ^2, I chose another relatively uninformed prior, such that $\sigma^2 \sim Exp(1/2)$. Therefore, I gave σ^2 a mean of 2 and a variance of 4. Both distrubtions are shown in Figure 1.

Figure 1: Prior Densities for μ and σ^2



Since this is not a conjugate model, I used a Gibbs sampler with a Gaussian random walk to estimate the parameters. However, since this can't be

displayed, the derivation of the posterior likelihood for both posterior distributions is shown below.

μ Posterior Likelihood

The Posterior Likelihood for the distribution of μ is, for $\mu \sim N(m, v^2)$ and $X|\mu, \sigma^2 \sim LogitN(\mu, \sigma^2)$,

$$f(\mu|X, \sigma^2) \propto f(\mu)L(X|\mu, \sigma^2)$$

We find $f(\mu)L(X|\mu, \sigma^2)$.

$$f(\mu)f(X|\mu, \sigma^2)$$

$$= (2\pi\sigma^2)^{-1/2} \exp\Big(-1/2\sigma^2\Big(\text{logit}(\mu) - m\Big)^2\Big)$$

$$*(2\pi\sigma^2)^{-n/2} \prod_{i=1}^{n} (x_i(1-x_i))^{-1} \exp \frac{-1}{2\sigma^2} \sum_{j=1}^{n} \Big(\text{logit}(x_j) - \mu\Big)^2$$

Notice that if we split this function up, we see the likelihood function used earlier to derive the MLE estimates. If we take the natrual logarithm of this function we get

$$l(X|\mu, \sigma^2) = -n/2\log(2\pi) - 1/2\log(\sigma^2) - \sum_{i=1}^{n}(x_i(1-x_i)) - 1/2\sigma^2 \sum_{j=1}^{n} \Big(\text{logit}(x_j) - \mu\Big)^2$$

If we take the natural logarithm of the Posterior Likelihood, we get

$$\log\Big(f(\mu)L(X|\mu, \sigma^2)\Big)$$

$$= (-n/2 - 1/2)\log(2\pi) - n/2\log(\sigma^2) - 1/2\log(v^2)$$

$$-n\log(v^2) - \sum_{i=1}^{n}(x_i(1-x_i)) - 1/2\sigma^2 \sum_{j=1}^{n} \Big(\text{logit}(x_j) - \mu\Big)^2 - 1/2/v^2\Big(\text{logit}(\mu) - m\Big)$$

$$\propto -n/2\log(\sigma^2) - 1/2\log(v^2) - 1/2\sigma^2 \sum_{j=1}^{n} \Big(\text{logit}(x_j) - \mu\Big)^2 - 1/2v^2\,\text{logit}(\mu) - m$$

I used this likelihood function for my Metroplis algorithm within my Gibbs sampler, substituting 0 and 2 for m and v^2 respectively.

σ^2 Posterior Likelihood

For the sake of not being repetitive, I'll use the log- likelihood function from the derivation before. Note that $\sigma^2 \sim Exp(\theta)$.

$$\log\Big(f(\sigma^2)L(X|\mu,\sigma^2)\Big)$$

$$= \log(\theta) - \sigma^2\theta$$

$$+(-n/2)\log(2\pi) - n/2\log(\sigma^2) - \sum_{i=1}^{n}(x_i(1-x_i)) - 1/2\sigma^2\sum_{j=1}^{n}\Big(\text{logit}(x_j) - \mu\Big)^2$$

$$\propto -n/2\log(\sigma^2) - 1/2\sigma^2\sum_{j=1}^{n}\Big(\text{logit}(x_j) - \mu\Big)^2 - \sigma^2\theta$$

This is the trimmed-down posterior likelihood that I used for my Metropolis algorithm in my Gibbs sampler, substituting 1/2 for θ.

4.2 Bayes II

My second estimator is extremely similar to the first, except that the shape parameter σ^2 is very heavily weighted around 2.

For my second set of priors, I claim $\mu \sim N(0,2)$, $\sigma^2 \sim IG(40,80)$. My prior for μ is identical. However, σ^2 is much more focused around 2. In comparing our Bayes II estimator with Bayes I and the MLE, we can see how a bad prior for σ^2 will affect μ and the functionality of the model.

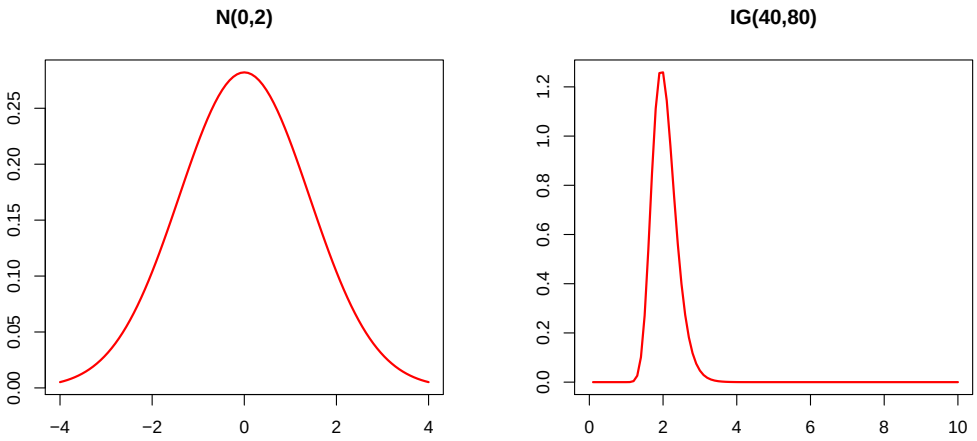
I chose the Inverse Gamma distribution because it is similar to the Exponential distribution but I can shrink the variance without changing the mean.

With the Exponential distribution, by contrast, every mean implies one and only one variance. Hence I can't create an Exponential distribution with mean 2 that has a variance less than 4. The Inverse Gamma distribution allows me that luxury.

To illustrate this concept, compare the prior distributions in Figure 2 with those in Figure 1.

Our Posterior Likelihood for μ is identical to that in Bayes I. Therefore I only show the derivation of the posterior likelihood for σ^2.

Figure 2: Prior Distributions for Bayes II



σ^2 Posterior Likelihood

Again, to spare any repetition, we use the log-likelihood derived earlier, such that

$$l(X|\mu,\sigma^2) = -n/2\log(2\pi) - 1/2\log(\sigma^2) - \sum_{i=1}^{n}(x_i(1-x_i)) - 1/2\sigma^2\sum_{j=1}^{n}\Big(\text{logit}(x_j) - \mu\Big)^2$$

Continuing with our derivation, observe that the PDF of the Inverse Gamma distribution is

$$f(\sigma^2) = \frac{\beta^\alpha}{\Gamma(\alpha)}(\sigma^2)^{-\alpha-1}\exp(-\beta/\sigma^2)$$

The natural logaritm of this is

$$\log(f(\sigma^2)) = l(\sigma^2) = \alpha\log(\beta) - \log(\Gamma(\alpha)) - (\alpha+1)\log(\sigma^2) - \frac{\beta}{\sigma^2}$$

We then see that

$$\log(f(\sigma^2)L(X|\mu,\sigma^2)) = l(\sigma^2) + l(X|\mu,\sigma^2)$$

$$= -n/2\log(2\pi) - 1/2\log(\sigma^2) - \sum_{i=1}^{n}(x_i(1-x_i)) - 1/2\sigma^2\sum_{j=1}^{n}\Big(\text{logit}(x_j) - \mu\Big)^2$$

$$+\alpha\log(\beta) - \log(\Gamma(\alpha)) - (\alpha+1)\log(\sigma^2) - \frac{\beta}{\sigma^2}$$

$$\propto -n/2\log(\sigma^2) - 1/2\sigma^2 \sum_{i=1}^{n} \Big(\text{logit}(x_i) - \mu\Big)^2 - (\alpha+1)\log(\sigma^2) - \frac{\beta}{\sigma^2}$$

This trimmed-likelihood function was used in my Metropolis Algorithm within my Gibbs sampler. Employing this derived likelihood, I obtained estimates of my parameters.

5 Application

The Logit-Normal models compositional or proportional data. Due to its relation with the normal distribution, I would suppose this is especially true if we have reason to suppose the data are distributed approximately normally.

I chose compositional data, describing the composition of milk. Specifically, I examined the fat composition of the data. Milk, when it is sold, is categorized according to fat composition. Also, herds and breeds on average contribute the same type of milk. In order to fit milk into the categories the stores use, producers "skim" the milk, churning it until the cream rises to the top and removing a certain amount.

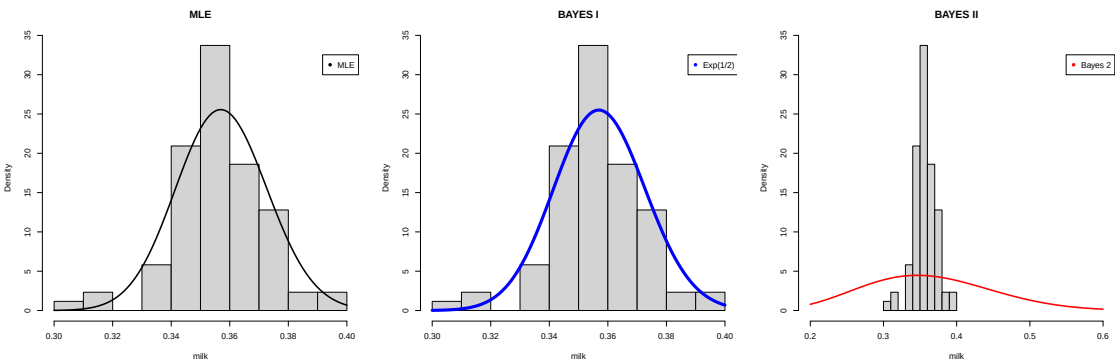
I wanted to model the fat composition of milk so we could determine how much cream to remove in order to fit sale specifications. We could then potentially remove a constant amount from each product and be assured it fits specifications within a certain probability.

The data is shown in a histogram below. I then superimposed my estimated distribution on top of the histogram.

We see that the MLE predicted model fits relatively well, although it would be preferable if we had more data.

My Bayes estimators performed quite well. Even my biased Bayes II estimator produced a similar distribution.

Figure 3: Data with Predicted Distributions



6 Confidence Interval Method

I used the Bootstrap Method in estimating my data. In order to do this I took a bootstrap sample and then used the generated sample for each of my estimators. I did this many times in order to generate a pseudo-distribution of the parameters.

Generating confidence intervals was then based on this generated data. I evaluated the likelihood function using my real data set and each bootstrap set of parameters. I then found the 5% quantile of my sample, and labeled those in the rejection region. Rejected sets of parameters for each of my three estimation methods are denoted below in red.

Also, I found confidence intervals on each of my parameters, as shown in the table below. For each of my estimation methods, I took the bootstrap samples of the estimates for each parameter and found the 2.5% and 97.5% quantiles. These then constitute 95% confidence intervals on each parameter for each estimation method.

	Estimate	Lower	Upper
Mu MLE	-0.59	-0.60	-0.58
S2 MLE	0.00	0.00	0.01
Mu B1	-0.59	-0.60	-0.58
S2 B1	0.00	0.00	0.01
Mu B2	-0.59	-0.60	-0.57
S2 B2	0.90	0.89	0.92

Table 1: Estimates and Confidence Intervals

Figure 4: Confidence Intervals and Regions for Our Data

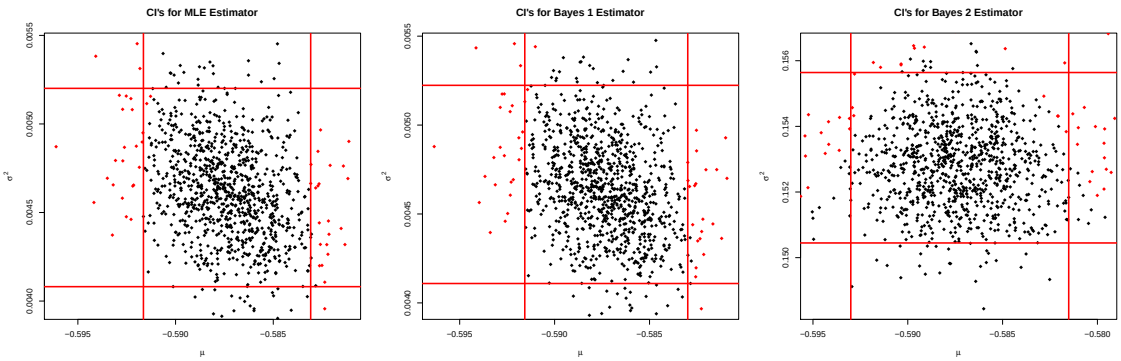
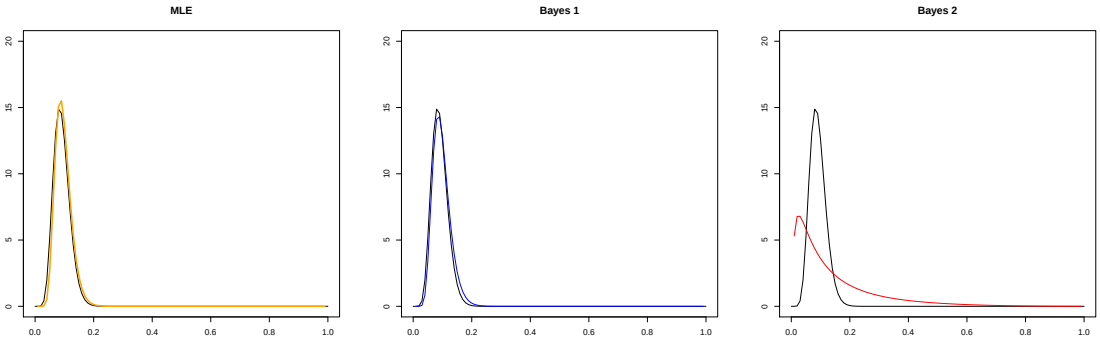


Figure 5: Comparison Of Predicted Distributions and Beta(10,100) Density



7 Model Misspecification

The logit normal can take many shapes, making it hard to determine a inappropriate distribution. However, I chose to generate data from a Beta distribution in order to break my Bayes estimators. Specifically, I chose to generate 30 draws from a Beta(10,100) distribution, which certainly won't be similar to a Logit-Normal random variable with a μ of 0 or a σ^2 of 2. This is in contrast to our prior estimates.

As you can see from the graph provided above, all three of my estimators actually gave decent estimates. The MLE guessed the distribution very well. My second Bayes estimator gave a estimate that is accurately centerd but incorrectly distributed. We see from this that the Logit-Normal is actually quite good at modelling to any distribution.

It appears that the Logit-Normal can model Beta data quite well. While this is true, we cannot find the mean, variance, or other moments of our distribution. Hence, if we can fit a Beta Distribution, it would be preferable to use the Beta rather than the Logit-Normal.

Despite all impediments, the Logit-Normal Distribution is a viable option for modelling data between 0 and 1. Applying our methods to real data in the former section showed it has its limitations. However, we are unsure how far of our estimators are from the true parameters for milk fat composition.

Figure 4 below denotes the true Beta distribution with a black line, while our proposed distributions are in other colors.

Figure 6: Bias: σ^2 Dependent

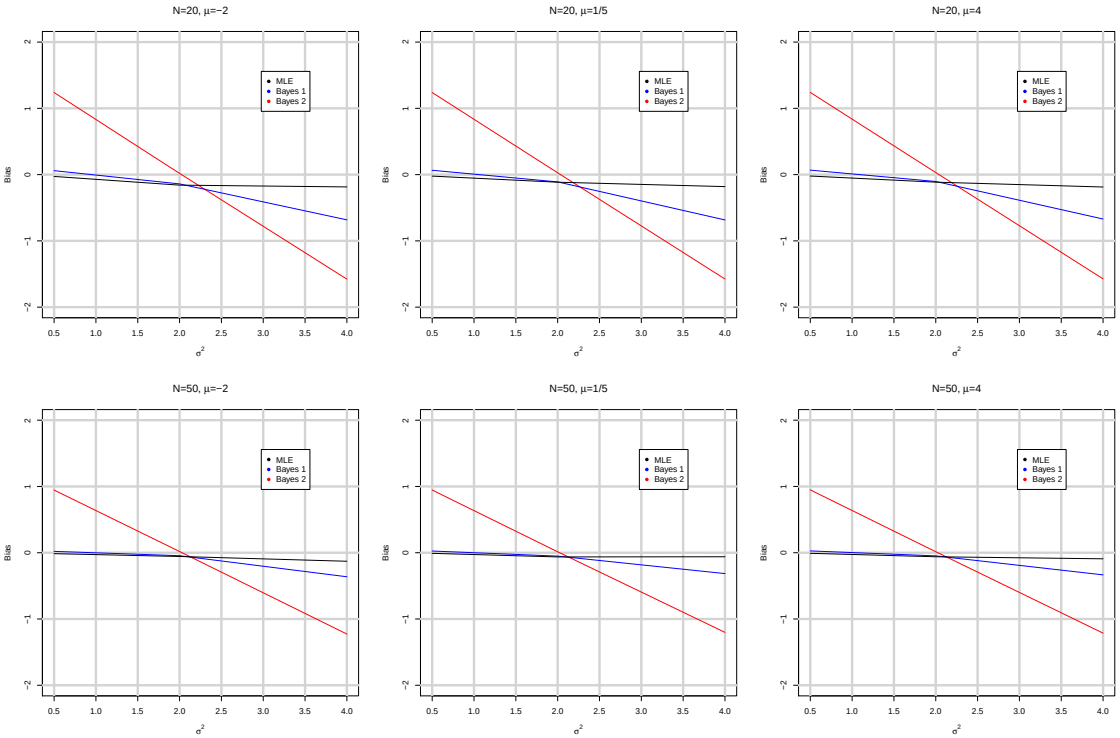


Figure 7: MSE: σ^2 Dependent

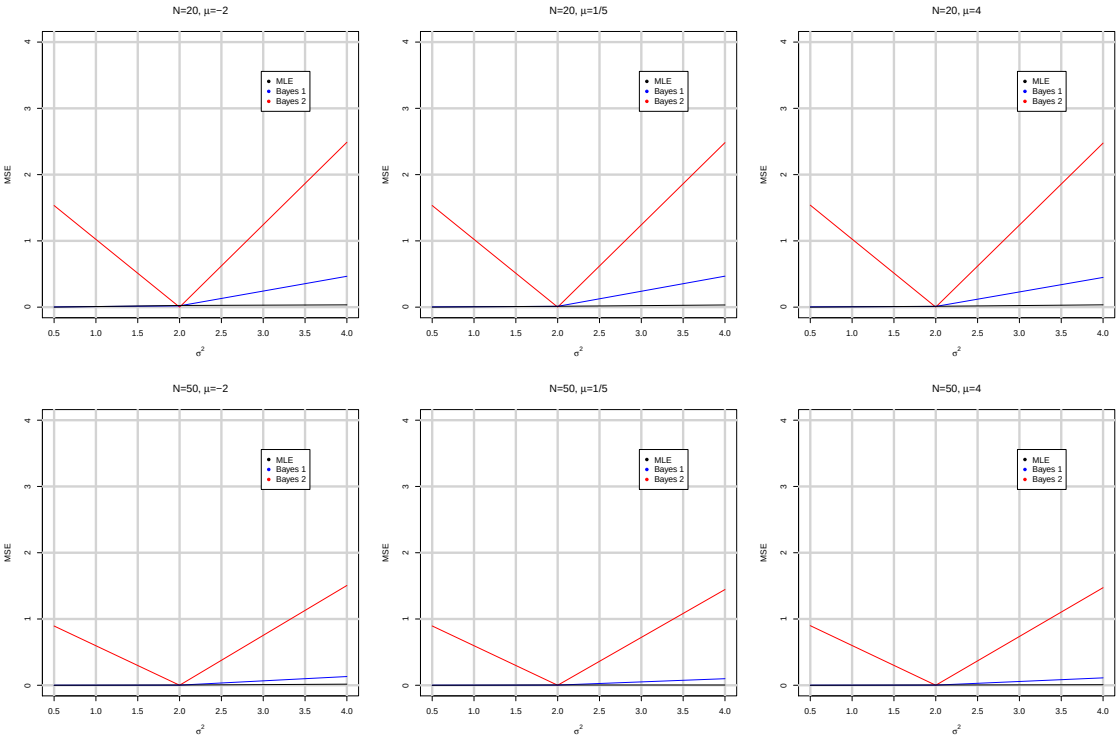


Figure 8: Bias: μ Dependent

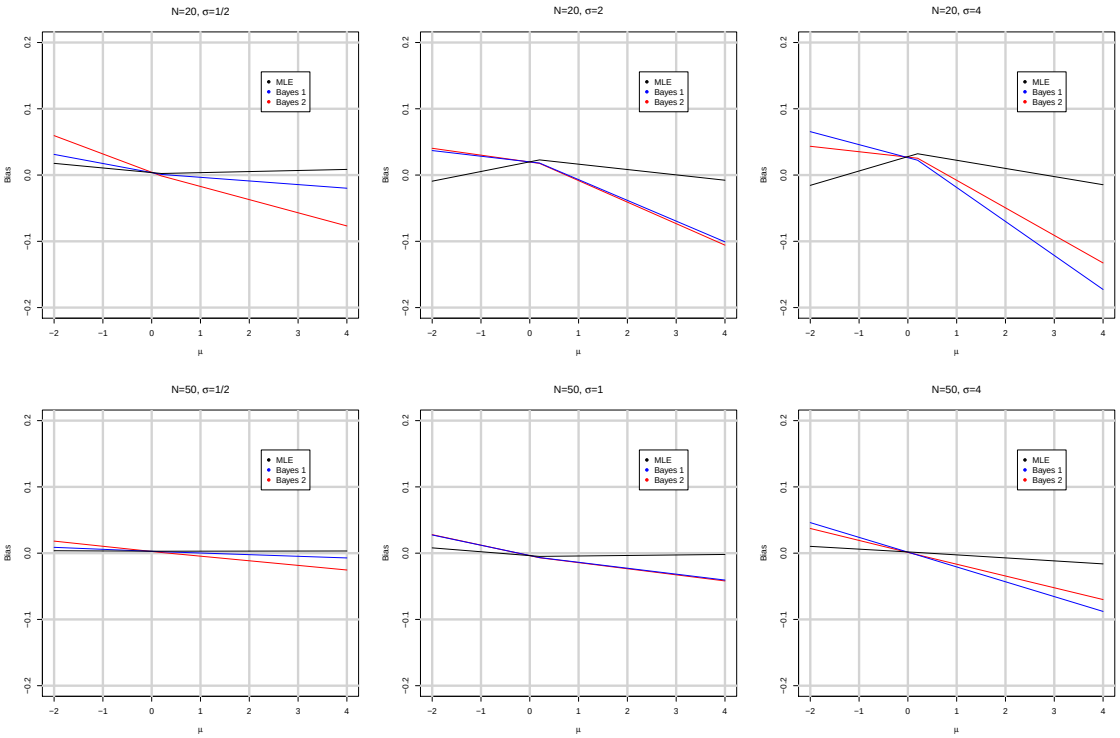
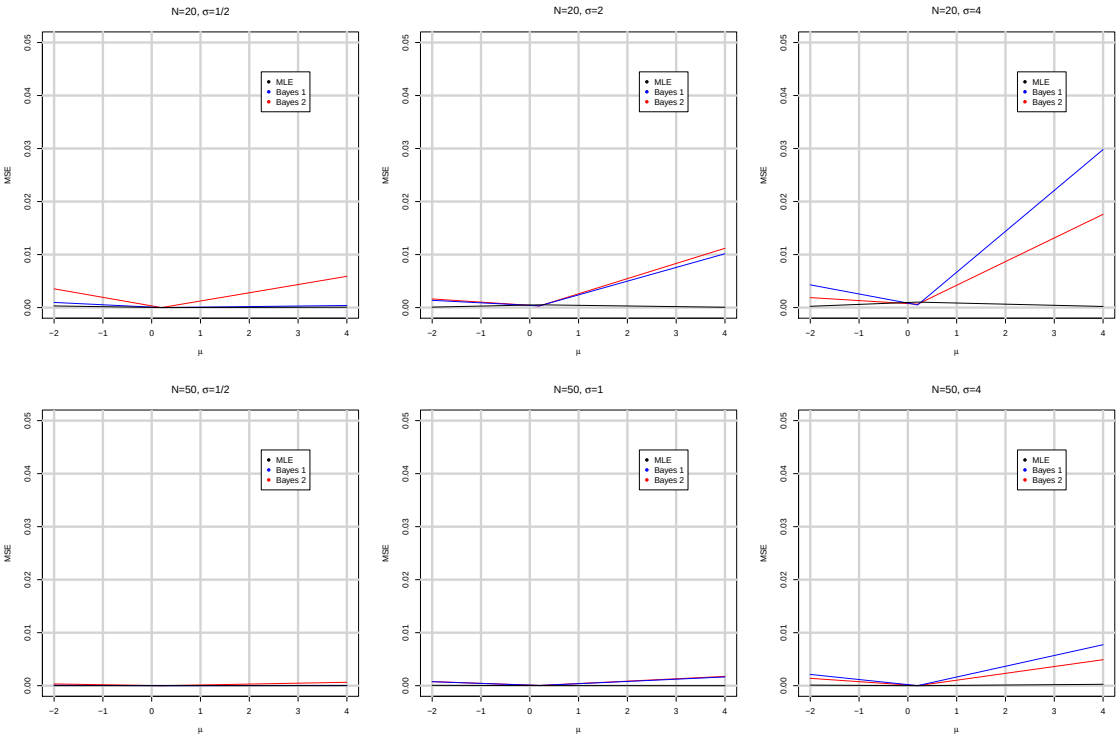


Figure 9: MSE: μ Dependent



8 Simulation Study

My simulation study was designed to test whether a bad prior on σ^2 would prompt poor estimation of μ. Also, I wanted to see how the Maximum Likelihood Estimators performed against my Bayes Estimators.

For this purpose, I chose 18 different treatments, comparing all possible combinations of the levels below:

$$N = 20 \text{ and } 50$$
$$\mu = -2,\ 1/5 \text{ and } 4.$$
$$\sigma^2 = 1/2,\ 2, \text{ and } 4.$$

Bias was then calculated using the true values for the simulated data. MSE was further calculated by squaring the bias measurements.

In order to make the results more readable, I held two of the levels constant and showed the change as the third parameter moves.

The data is presented in the Appendix in Figures 6-7. The graphs are all standardized within each figure, such that the scale for the x and y axes are the same within figures for comparability.

8.1 μ

First we consider the case where σ^2 is the dependent variable. Our primary question of interest is well does Bayes I estimate σ^2 against the MLE estimates.

We see from Figures 6-7 that the MLE trounces the other two estimators in terms of bias and MSE. While this is not altogether unexpected, it is diappointing that the Bayes I estimator didn't outperform the MLE even when the true σ^2 was equal to the prior estimate.

The Bayes II estimator performed badly. This was altogether expected. In it, we can see the trend followed by both Bayes estimators where the bias and MSE approach 0 as the true value for σ^2 approaches the prior estimate for σ^2.

8.2 σ^2

We chose our Bayesian estimators in order to see the effect σ^2 has on the estimates for μ. Specifically, we were looking divergence between the red and blue lines.

As can be seen in Figures 8-9, the expected divergence did not happen. The prior for σ^2 had no distinct effect on our estimates for μ.

While this was disheartening, we see that the MLE still greatly outperforms the Bayesian Estimators. The bias and MSE when $\sigma^2 = 2$, $N = 20$ show that the Bayesian estimates are slighly biased when $\hat{\mu}$ is close to the true value of μ.

9 Summary

After evaluating the Logit-Normal distribution, I'm quite surprised it isn't used more frequently. Through our simulation study, along with our misclassification and real-world application studies, showed that the Maximum Likelihood Estimates for each of our parameter values did very well.

After consideration, I think the Logit-Normal is used infrequently because it has no moments. The Beta distribution covers the same support, and yet has moments, a closed-form CDF, and the ability to flexibly model many different arrangements.

The Logit-Normal therefore falls by the wayside. Our search for the best model to make estimates on data is by necessity Darwinian. Since the Logit-Normal isn't as advantageous due to its lack of moments, other distributions such as the Beta are used.

Our study implies further fields of research. While bias and MSE estimates denote abnormalities, denoted above, a study into the difference in capability between the Beta and the Logit-Normal Distribution would be advisable. Such a study might provide a niche where the Logit-Normal outperforms all other proposed distributions. Further applications could then be found for this pliable, accurate distribution which will open up new fronteirs of data analysis.

10 Appendix

10.1 References

- Aitchinson, J.(1980) Logistic-Normal Distributions: Some Properties and Uses, *Biometrika*,Vol. 67, No. 2, pp. 261-272

- Daudin, J.J. Duby, C. and Trecourt, P. (1988) Stability of Principal Component Analysis Studied by the Bootstrap Method; *Statistics* 19, 241258.

10.2 Additional Figures and Tables

Below are the estimates for each parameter for each combination of levels. I included a 95% Confidence Interval for each estimate to account for Monte Carlo error.

Also included below are the figures mentioned in the Simulation Study section.

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-1.98	-1.99	-1.97
MLE: S2	0.47	0.46	0.48
B1: Mu	-1.97	-1.98	-1.96
B1: S2	0.56	0.55	0.57
B2: Mu	-1.94	-1.95	-1.93
B2: S2	1.74	1.74	1.74

Table 2: N=20, σ^2=0.5, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-2.01	-2.03	-1.99
MLE: S2	1.84	1.81	1.88
B1: Mu	-1.96	-1.98	-1.94
B1: S2	1.86	1.83	1.89
B2: Mu	-1.96	-1.98	-1.94
B2: S2	2.02	2.01	2.03

Table 3: N=20, σ^2=2, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-2.02	-2.04	-1.99
MLE: S2	3.81	3.74	3.89
B1: Mu	-1.93	-1.96	-1.91
B1: S2	3.32	3.27	3.37
B2: Mu	-1.96	-1.98	-1.93
B2: S2	2.42	2.41	2.44

Table 4: N=20, σ^2=4, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.20	0.19	0.21
MLE: S2	0.48	0.47	0.49
B1: Mu	0.20	0.19	0.21
B1: S2	0.57	0.55	0.58
B2: Mu	0.20	0.19	0.21
B2: S2	1.74	1.74	1.74

Table 5: N=20, σ^2=0.5, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.22	0.20	0.24
MLE: S2	1.89	1.85	1.92
B1: Mu	0.22	0.20	0.24
B1: S2	1.89	1.86	1.92
B2: Mu	0.22	0.20	0.24
B2: S2	2.03	2.02	2.04

Table 6: N=20, σ^2=2, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.23	0.20	0.26
MLE: S2	3.82	3.74	3.89
B1: Mu	0.22	0.20	0.25
B1: S2	3.32	3.27	3.37
B2: Mu	0.23	0.20	0.25
B2: S2	2.42	2.41	2.44

Table 7: N=20, σ^2=4, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	4.01	4.00	4.02
MLE: S2	0.48	0.47	0.49
B1: Mu	3.98	3.97	3.99
B1: S2	0.57	0.56	0.58
B2: Mu	3.92	3.91	3.93
B2: S2	1.74	1.74	1.74

Table 8: N=20, σ^2=0.5, μ=4

	Estimate	Lower Bound	Upper Bound
MLE: Mu	3.99	3.97	4.01
MLE: S2	1.89	1.85	1.93
B1: Mu	3.90	3.88	3.92
B1: S2	1.90	1.86	1.93
B2: Mu	3.89	3.87	3.91
B2: S2	2.03	2.02	2.04

Table 9: N=20, σ^2=2, μ=4

	Estimate	Lower Bound	Upper Bound
MLE: Mu	3.99	3.96	4.01
MLE: S2	3.81	3.74	3.89
B1: Mu	3.83	3.80	3.85
B1: S2	3.33	3.28	3.38
B2: Mu	3.87	3.84	3.90
B2: S2	2.43	2.41	2.44

Table 10: N=20, σ^2=4, μ=4

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-2.00	-2.00	-1.99
MLE: S2	0.49	0.48	0.49
B1: Mu	-1.99	-2.00	-1.98
B1: S2	0.52	0.51	0.53
B2: Mu	-1.98	-1.99	-1.98
B2: S2	1.45	1.44	1.45

Table 11: N=50, σ^2=0.5, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-1.99	-2.00	-1.98
MLE: S2	1.94	1.92	1.97
B1: Mu	-1.97	-1.98	-1.96
B1: S2	1.95	1.93	1.98
B2: Mu	-1.97	-1.98	-1.96
B2: S2	2.02	2.01	2.03

Table 12: N=50, σ^2=2, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	-1.99	-2.01	-1.97
MLE: S2	3.87	3.82	3.92
B1: Mu	-1.95	-1.97	-1.94
B1: S2	3.64	3.60	3.68
B2: Mu	-1.96	-1.98	-1.95
B2: S2	2.77	2.75	2.79

Table 13: N=50, σ^2=4, μ=-2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.20	0.20	0.21
MLE: S2	0.49	0.48	0.50
B1: Mu	0.20	0.20	0.21
B1: S2	0.53	0.52	0.53
B2: Mu	0.20	0.19	0.21
B2: S2	1.45	1.44	1.45

Table 14: N=50, σ^2=0.5, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.20	0.18	0.21
MLE: S2	1.94	1.91	1.96
B1: Mu	0.19	0.18	0.21
B1: S2	1.95	1.93	1.97
B2: Mu	0.19	0.18	0.21
B2: S2	2.01	2.00	2.02

Table 15: N=50, σ^2=2, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	0.20	0.18	0.22
MLE: S2	3.94	3.89	3.99
B1: Mu	0.20	0.18	0.21
B1: S2	3.69	3.64	3.73
B2: Mu	0.20	0.18	0.22
B2: S2	2.80	2.78	2.82

Table 16: N=50, σ^2=4, μ=0.2

	Estimate	Lower Bound	Upper Bound
MLE: Mu	4.00	4.00	4.01
MLE: S2	0.49	0.49	0.50
B1: Mu	3.99	3.99	4.00
B1: S2	0.53	0.52	0.53
B2: Mu	3.97	3.97	3.98
B2: S2	1.45	1.45	1.45

Table 17: N=50, σ^2=0.5, μ=4

	Estimate	Lower Bound	Upper Bound
MLE: Mu	4.00	3.99	4.01
MLE: S2	1.94	1.91	1.96
B1: Mu	3.96	3.95	3.97
B1: S2	1.95	1.93	1.98
B2: Mu	3.96	3.95	3.97
B2: S2	2.02	2.01	2.02

Table 18: N=50, σ^2=2, μ=4

	Estimate	Lower Bound	Upper Bound
MLE: Mu	3.98	3.97	4.00
MLE: S2	3.91	3.86	3.96
B1: Mu	3.91	3.89	3.93
B1: S2	3.67	3.63	3.71
B2: Mu	3.93	3.91	3.95
B2: S2	2.79	2.77	2.81

Table 19: N=50,σ^2=4, μ=4